Confusion matrix

A confusion matrix is like a scorecard for a computer program that predicts things. It shows how many times the program got things right (true positives and true negatives) and how many times it made mistakes (false positives and false negatives). This helps us understand how well the program is doing and how to make it better.

A confusion matrix is essential for assessing a classification model's performance, as it breaks down predictions into true positives, true negatives, false positives, and false negatives, enabling the calculation of various evaluation metrics. It helps in error analysis, model improvement, and class imbalance detection.

In short

A confusion matrix is like a tool to see how often a computer prediction is correct and where it's making mistakes.

Case Study : Quality Control in a Manufacturing Process

Now, you create a confusion matrix to summarize the model's performance:

Cases 1000	**Predicted Defective**	**Predicted Non-Defective**
Actual Defective	90 (TP)	10 (FN)
Actual Non-Defective	20 (FP)	880 (TN)

0.97 % Accuracy

Analysis of the Confusion Matrix

True Positives (TP): These are the defective products correctly classified as defective. In this case, 90 defective products were correctly identified.

True Negatives (TN): These are the non-defective products correctly classified as non-defective. Here, 880 non-defective products were correctly identified.

False Positives (FP): These are the non-defective products incorrectly classified as defective. In this example, 20 non-defective products were mistakenly identified as defective.

False Negatives (FN): These are the defective products incorrectly classified as non-defective. In this case, 10 defective products were missed and classified as non-defective.

Case Study : Medical Test for a Rare Disease

After making predictions, you create a confusion matrix to summarize the model's performance:

0.95 % Accuracy

Cases 500	**Predicted Positive (Has Disease X)**	**Predicted Negative (Does Not Have Disease X)**
Actual Positive	40 (True Positives)	10 (False Negatives)
Actual Negative	15 (False Positives)	435 (True Negatives)

Analysis of the Confusion Matrix

True Positives (TP): These are the patients with Disease X correctly classified as having the disease. In this case, 40 patients with the disease were correctly identified.

True Negatives (TN): These are the patients without Disease X correctly classified as not having the disease. Here, 435 patients without the disease were correctly identified.

False Positives (FP): These are the patients without the disease incorrectly classified as having the disease. In this example, 15 patients without the disease were mistakenly identified as having the disease.

False Negatives (FN): These are the patients with Disease X incorrectly classified as not having the disease. In this case, 10 patients with the disease were missed and classified as not having it.

Case Study : Credit Card Fraud Detection

After model predictions, you construct a confusion matrix:

Cases 2000	Predicted Fraudulent	Predicted Legitimate
Actual Fraudulent	45 (True Positives)	5 (False Negatives)
Actual Legitimate	10 (False Positives)	1,940 (True Negatives)

0.99% accuracy

i. ***True Positives (TP):*** These are the fraudulent transactions correctly classified as fraudulent. In this case, **45 fraudulent transactions were correctly identified.**
ii. ***True Negatives (TN):*** These are the legitimate transactions correctly classified as legitimate. Here, **1,940 legitimate transactions were correctly identified.**
iii. ***False Positives (FP):*** These are the legitimate transactions incorrectly classified as fraudulent. In this example, **10 legitimate transactions were mistakenly classified as fraudulent.**
iv. ***False Negatives (FN):*** These are the fraudulent transactions incorrectly classified as legitimate. In this case, **5 fraudulent transactions were missed and classified as legitimate.**

	Predicted		
165	No	Yes	
Actual No	50	10	60
Actual Yes	5	100	105
	55	110	

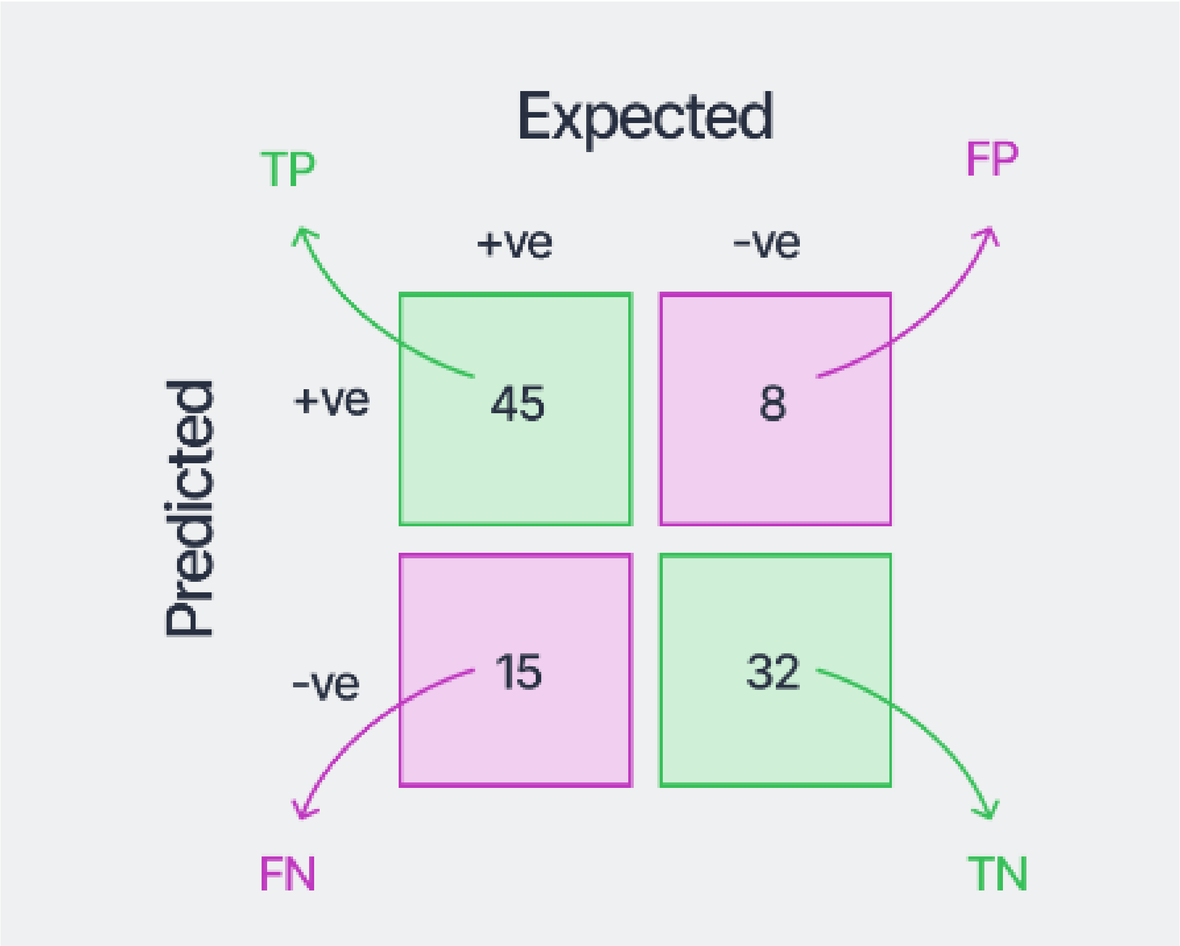
	Predicted		
165	No	Yes	
Actual No	50 [TN]	10 [FP]	60
Actual Yes	5 [FN]	100 [TP]	105
	55	110	

Example

Suppose we have a binary class imbalanced dataset consisting of 60 samples in the positive class and 40 samples in the negative class of the test set, which we use to evaluate a machine learning model.

- *45 samples* belonging to the positive class being predicted correctly.
- *32* sample belonging to the negative class being predicted correctly.
- *8* sample belonging to the negative class but being predicted wrongly as belonging to the positive class.
- 15 sample belonging to the positive class but being predicted wrongly as belonging to the negative class.
- Create confusion metrics, calculate accuracy, precision, recall and F1 score.

Solution:



$$\text{Accuracy} = \frac{TP + TN}{TP + FP + TN + FN}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$\text{F1-Score} = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

	Predicted		
165	No	Yes	
Actual No	50 [TN]	10 [FP]	60
Actual Yes	5 [FN]	100 [TP]	105
	55	110	

Accuracy = (TP +TN)/Total

= (100 + 50)/165

= 0.91

Accuracy

Accuracy measures the proportion of correct predictions.

Accuracy = (True Positives + True Negatives) / Total Number of Predictions

---

Error rate = 1- Accuracy

= 1- 0.91

= 0.09

Error rate

The error rate measures the proportion of incorrect predictions.

Error Rate = (False Positives + False Negatives) / Total Number of Predictions

OR

= (FP + FN) /Total

= (10+5)/165

= 0.09

MODEL A	Predicted		
165	No	Yes	
Actual SPAM No	50 [TN]	10 [FP]	60
Actual SPAM Yes	5 [FN]	100 [TP]	105
	55	110	

MODEL B	Predicted		
165	No	Yes	
Actual SPAM No	50 [TN]	04 [FP]	54
Actual SPAM Yes	11 [FN]	100 [TP]	111
	61	104	

Which model you will prefer either model A or Model B Considering false positives?

Precision

Model A

Precision = (True Positives) / (True Positives + False Positives)

= (100)/ (100+10)

=0.90

Model B

Precision = (True Positives) / (True Positives + False Positives)

= (100)/ (100+4)

=0.96

Precision is important when false positives (Type 1 Error) are more concerned.

Precision

It tells you how many of the predicted positive cases were actually correct.

Precision = (True Positives) / (True Positives + False Positives)

Model A 10000 Patients		Predicted: Detected Cancer	Predicted: Not Detected Cancer
Actual	Has Cancer	1000 TP	200 FN
	No Cancer	800 FP	8000 TN

Model B 10000 Patients	Predicted: Detected Cancer	Predicted: Not Detected Cancer
Has Cancer	1000 TP	500 FN
No Cancer	500 FP	8000 TN

Which model you will prefer either model A or Model B Considering false negatives?

Recall

Model A

= Recall = (True Positives) / (True Positives + False Negatives)

=1000/ (1000+200)

= 0.83

Model B

= Recall = (True Positives) / (True Positives + False Negatives)

=1000/ (1000+500)

= 0.66

Recall is important when false negatives (Type 2 error) are more concerned.

Recall

Recall tells you how many of the actual positive cases were correctly predicted by the model.

Recall = (True Positives) / (True Positives + False Negatives)

165	Predicted		
	No	Yes	
Actual No	50 [TN]	10 [FP]	60
Actual Yes	5 [FN]	100 [TP]	105
	55	100	

Precision

= 100/ (100+10)

= 100/110

= 0.90

Recall

= 100/ (100+5)

= 100/105

= 0.95

F1 score

F1 Score = 2 * (Precision * Recall) / (Precision + Recall)

F1 Score = 2* (0.90 * 0.95)/ (0.90 + 0.95)

F1 Score = 0.91

The F1 score is a metric that combines precision and recall to assess the accuracy of classification models. It provides a balanced measure of a model's ability to minimize both false positives and false negatives.

F1 Score = 2 * (Precision * Recall) / (Precision + Recall)

Multiclass Confusion Matrix

Predicted

Actual

Samples = 110	Apple	Mango	Guava	Total
Apple	25	5	7	37
Mango	4	32	2	38
Guava	6	4	25	35
Total	35	41	34	110